Transistors and Logic Gates

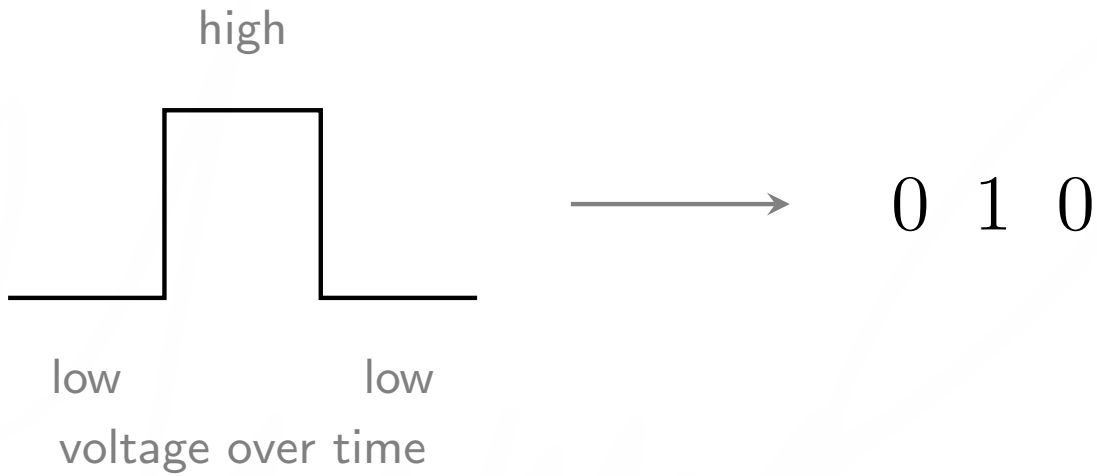
Computer Organization

Chandra Gummaluru

chandra-gummaluru.github.io

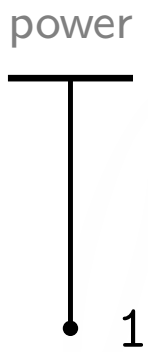


Recall that we break up the voltage range into two levels.



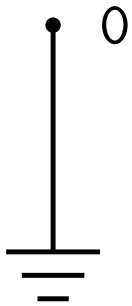
a voltage above some upper threshold is read as a 1, and a voltage below a lower threshold is read as a 0

A wire can carry one of these two values, depending on what it is tied to.



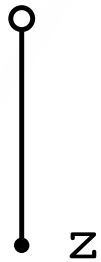
connected to power

carries a 1



connected to ground

carries a 0



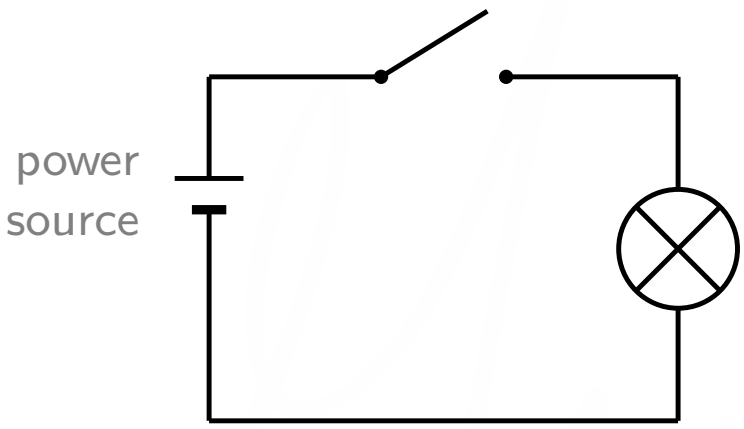
connected to nothing

carries an undefined value, z

However, its value can also be undefined if it is not tied to anything (this is different from carrying a 0).

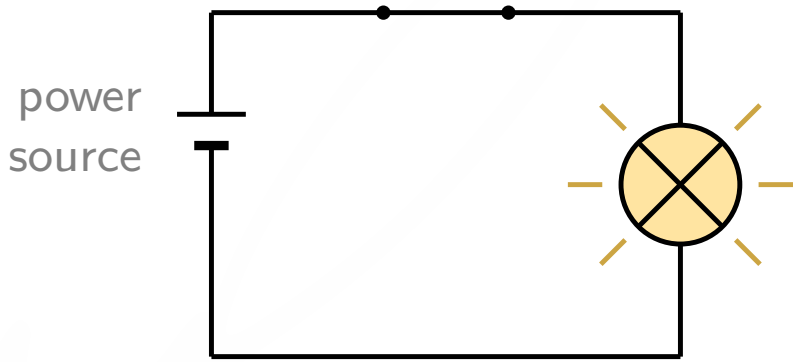
Switches allow us to change what a wire is tied to.

open



the switch breaks the loop,
so the bulb stays off

closed

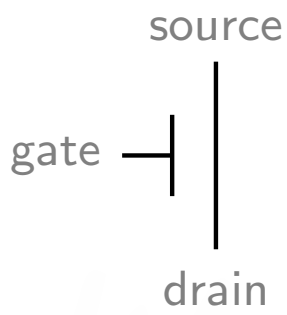


the switch completes the loop,
so the bulb turns on

To build a computer, we need to create a switch that can be moved by sending an electrical signal.

the transistor

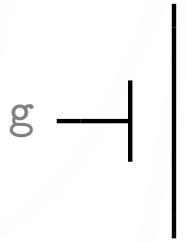
A transistor has three terminals. The voltage on the **gate** decides whether current may flow between the **source** and the **drain**.



the gate acts like the handle of a switch

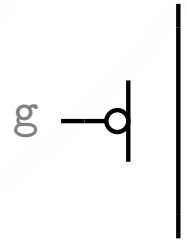
Two kinds of transistor respond to the gate in opposite ways.

nMOS



conducts when the gate is 1

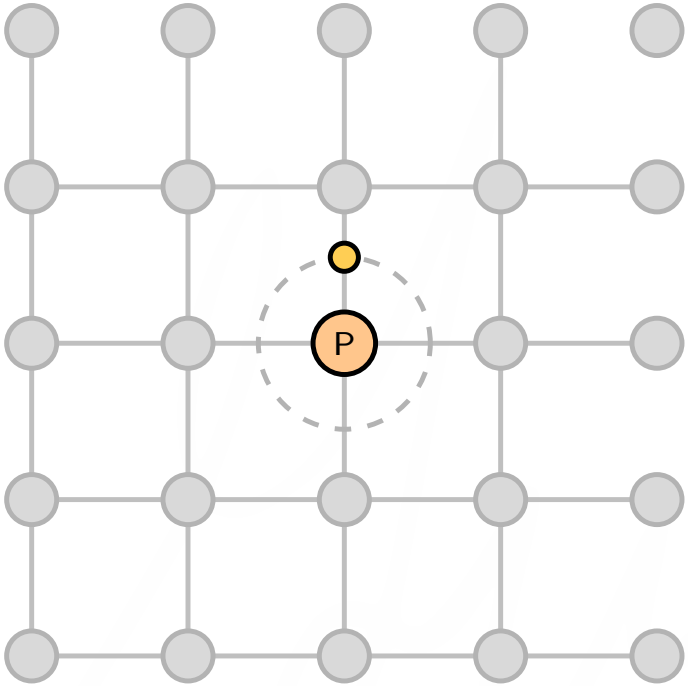
pMOS



conducts when the gate is 0

Transistors use a silicon base, but pure silicon has no free charges.

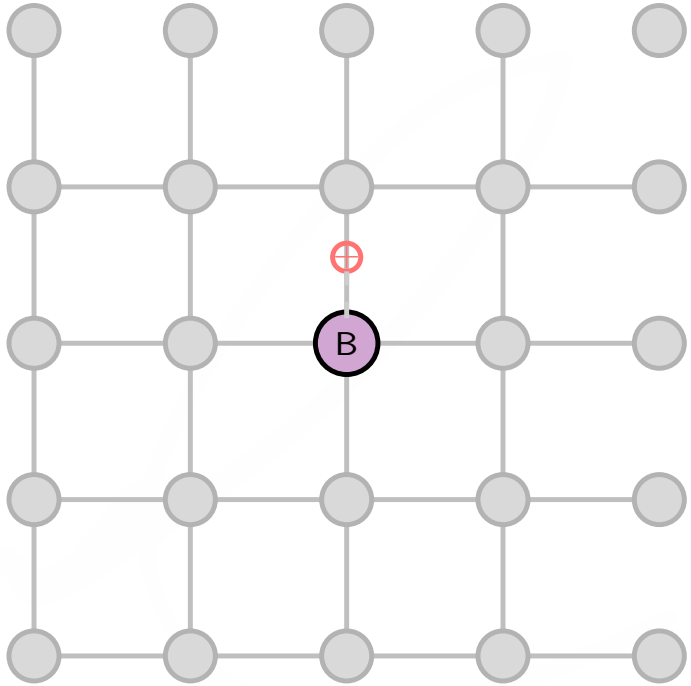
n-type



phosphorus donates a spare electron

extra electrons are free to carry ($-$) charge

p-type



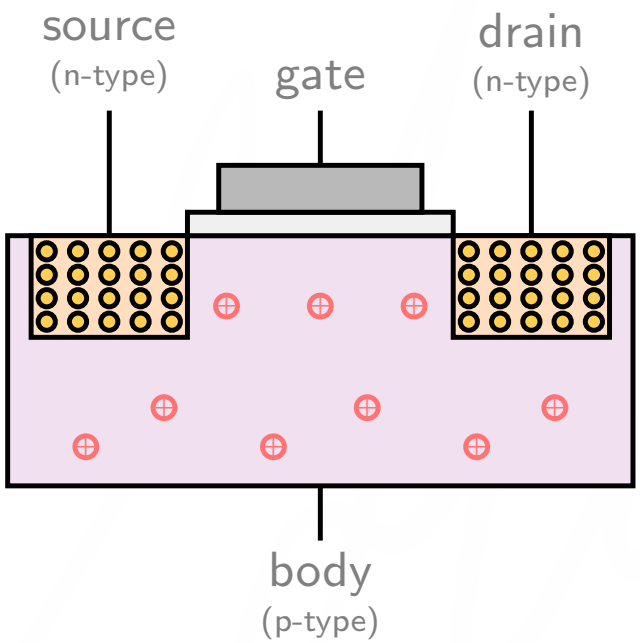
boron leaves a missing bond (a hole)

extra holes are free to carry ($+$) charge

Adding a small amount of another element (called **doping**), creates extra electrons or extra holes (lack of electrons).

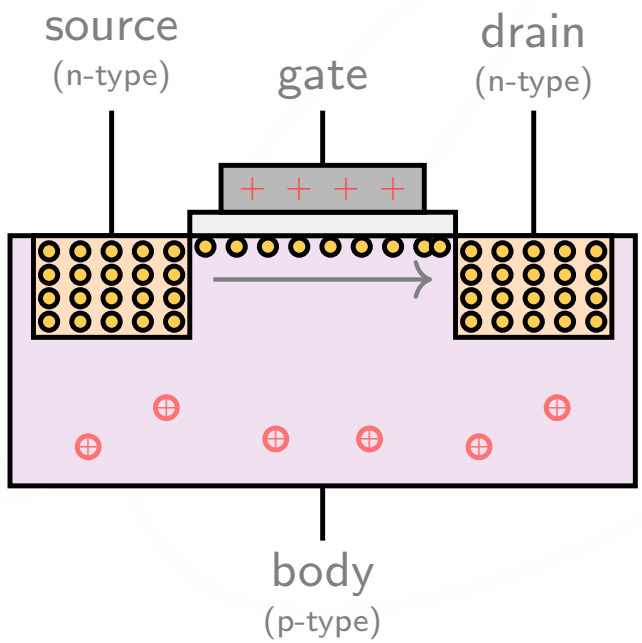
The source and the drain are made of one type and the base is made of another. The gate is separated from the base by an insulating layer.

gate off $(V_G = 0)$



few free electrons in the base
disturbance cannot move across base

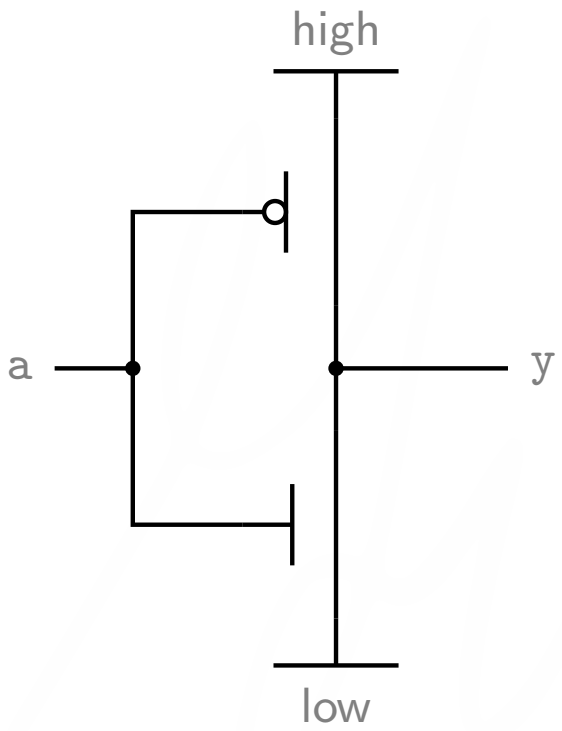
gate on $(V_G > V_{th})$

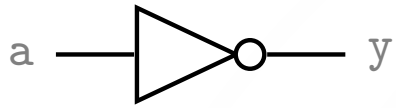


electrons drawn from source and drain
disturbance can move across base

The gate voltage controls whether electrons connect the source and drain.

NOT gate

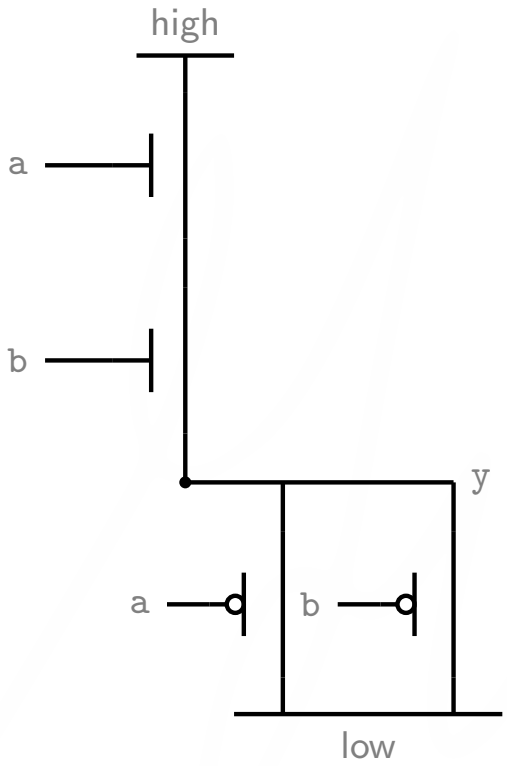


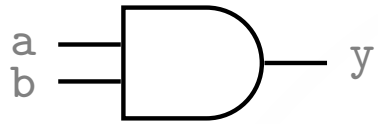


a	y
0	1
1	0

the transistor circuit, circuit symbol, and truth table for the NOT gate

AND gate

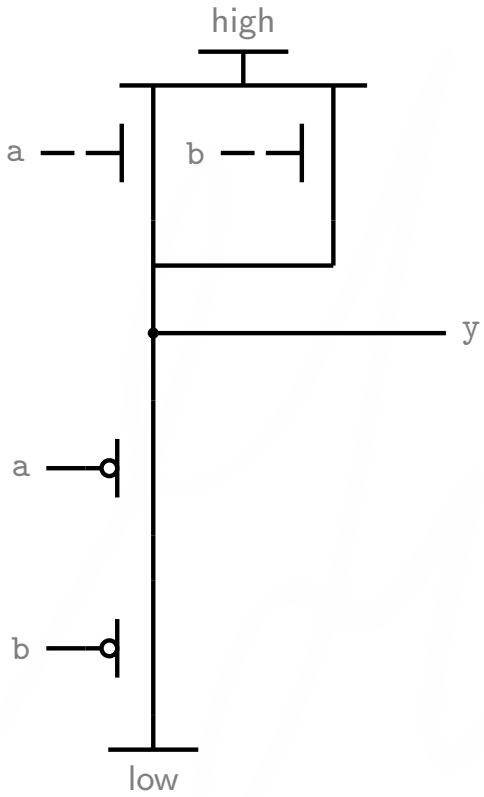


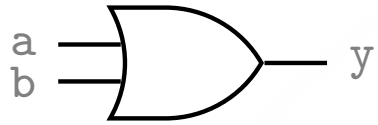


a	b	y
0	0	0
0	1	0
1	0	0
1	1	1

the transistor circuit, circuit symbol, and truth table for the AND gate

OR gate



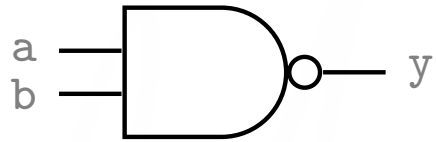


a	b	y
0	0	0
0	1	1
1	0	1
1	1	1

the transistor circuit, circuit symbol, and truth table for the OR gate

Adding a NOT gate to the output of an AND gate or OR gate gives two more gates (a NAND and a NOR, respectively), drawn with a circle on the output.

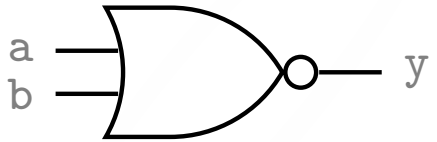
NAND gate



1 unless both inputs are 1

a	b	y
0	0	1
0	1	1
1	0	1
1	1	0

NOR gate

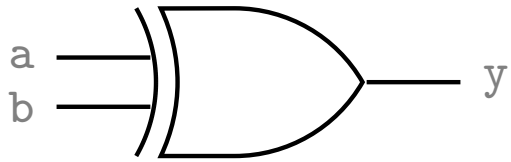


1 only when both inputs are 0

a	b	y
0	0	1
0	1	0
1	0	0
1	1	0

XOR gate

We can build an exclusive OR gate (XOR gate) by combining an OR gate and a NAND gate.



a	b	y
0	0	0
0	1	1
1	0	1
1	1	0

output is 1 when the inputs differ

These few gates are enough to build every circuit in the rest of the course.